# Medium Office Network Documentation

*Cisco Packet Tracer lab project*

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I built this office network to practise connecting multiple departments while controlling which services each group could access. I wanted a working internal network, guest internet access and a public website in a separate DMZ, with an admin PC used to manage the network devices.

I tested DHCP, DNS, website access, PAT and SSH as I built the network. Once those worked, I added restrictions and retested from each department to check that the right connections were allowed or blocked.

## What I built

The scenario represents an office of around 40 to 60 employees, using a smaller set of PCs to test each role. I used two routers, five switches, four servers, nine PCs and two printers. The ISP and internet website were simulated inside Packet Tracer.

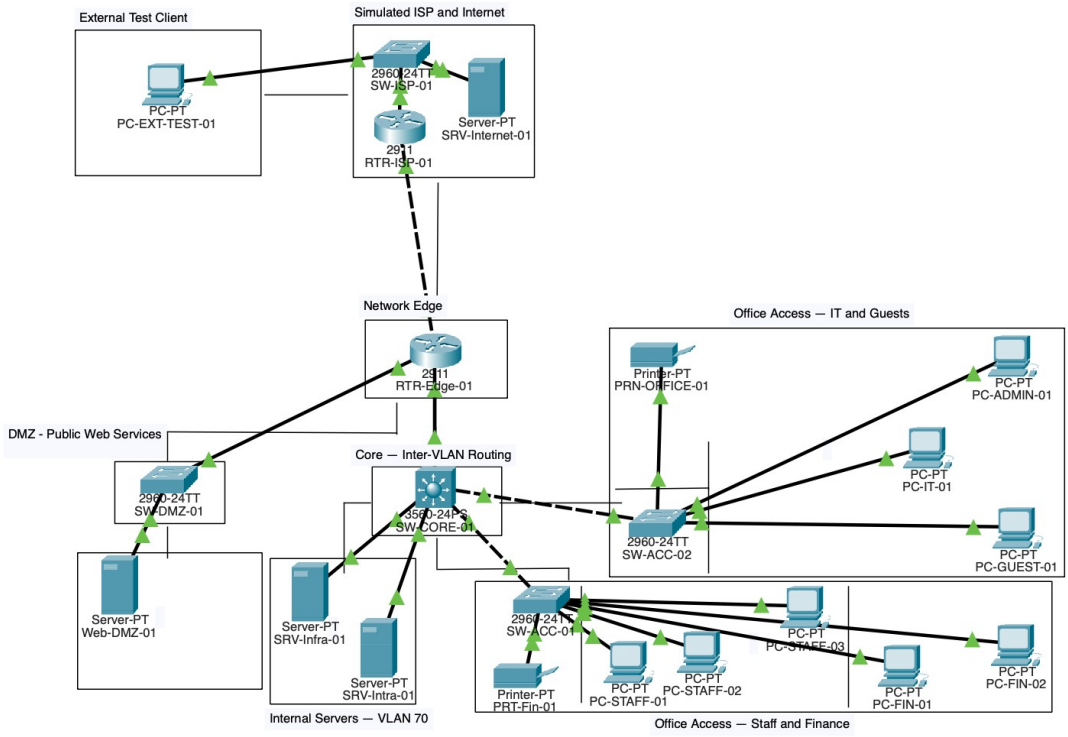



**Figure 1 Final Packet Tracer topology showing the external network, edge, DMZ, core, internal servers and office access areas.**

## How I approached the build

I worked through the ISP and DMZ first, then the internal VLANs and services. After basic connectivity worked, I added PAT, ACLs and SSH management. I saved screenshots and running configurations throughout the build.

# Devices and physical connections

I used a multilayer switch for the core because it needed to route between VLANs. The 2960 access switches handled endpoint connections and carried the required VLANs back to the core over trunks.

| Device or group | Model | Purpose |
|---|---|---|
| RTR-ISP-01 and RTR-Edge-01 | 2 × Cisco 2911 | Simulated ISP and office perimeter |
| SW-CORE-01 | 1 × Catalyst 3560 | Internal routing, gateways and ACLs |
| SW-ACC-01 and SW-ACC-02 | 2 × Catalyst 2960 | Office endpoint access |
| SW-DMZ-01 and SW-ISP-01 | 2 × Catalyst 2960 | DMZ and simulated external LAN |
| SRV-Infra-01 and SRV-Intra-01 | 2 × Server-PT | Internal DHCP/DNS and intranet |
| WEB-DMZ-01 and SRV-Internet-01 | 2 × Server-PT | Public website and external DNS/web |
| Staff 3, Finance 2, IT 1, admin 1, guest 1, external 1 | 9 × PC-PT | Role-based connectivity tests |
| Office and Finance printers | 2 × Printer-PT | Shared resource access tests |

## Port allocation

| Device and port | Connection or role | VLAN or mode |
|---|---|---|
| Edge G0/0 | ISP G0/0 | Routed WAN |
| Edge G0/1 | DMZ switch G0/1 | DMZ gateway |
| Edge G0/2 | Core G0/1 | Routed transit |
| Core F0/23 | ACC-01 G0/1 | Trunk 10, 20, 40, 99 |
| Core F0/24 | ACC-02 G0/1 | Trunk 30, 40, 60, 99 |
| Core F0/1 and F0/2 | Infrastructure and intranet servers | Access 70 |
| ACC-01 F0/1 to F0/3 | Staff PCs | Access 10 |
| ACC-01 F0/13 and F0/14 | Finance PCs | Access 20 |
| ACC-01 F0/21 | Finance printer | Access 40 |
| ACC-02 F0/1 and F0/2 | IT PC and admin PC | Access 30 |
| ACC-02 F0/5 and F0/9 | Office printer and guest PC | Access 40 and 60 |
| DMZ switch G0/1 and F0/1 | Edge and public web server | Access 50 |
| ISP switch G0/1, F0/1, F0/2 | ISP router, server, external PC | Access 1 |

I used straight-through cables for endpoints and routers connected to switches, crossover cables between switches and between the two routers, and console cables for local configuration.

# IPv4 and VLAN addressing

I used a separate /24 subnet for each internal group and the DMZ. The VLAN number matches the third octet of the internal subnet to make addresses easier to follow. The two router transit networks use /30 subnets.

| VLAN | Purpose | Subnet | Gateway |
|---|---|---|---|
| 10 | Staff | 10.10.10.0/24 | 10.10.10.1 |
| 20 | Finance | 10.10.20.0/24 | 10.10.20.1 |
| 30 | IT and admin | 10.10.30.0/24 | 10.10.30.1 |
| 40 | Printers | 10.10.40.0/24 | 10.10.40.1 |
| 50 | DMZ | 172.16.50.0/24 | 172.16.50.1 |
| 60 | Guest | 10.10.60.0/24 | 10.10.60.1 |
| 70 | Internal servers | 10.10.70.0/24 | 10.10.70.1 |
| 99 | Management | 10.10.99.0/24 | 10.10.99.1 |
| 999 | Unused office ports | No subnet | No gateway |

The core owns the internal gateways. The edge router owns the DMZ gateway. VLAN 50 stays on the dedicated DMZ switch and is not carried on the office trunks.

### Transit and external addresses

| Device interface | Address | Connection |
|---|---|---|
| RTR-ISP-01 G0/0 | 203.0.113.1/30 | Edge G0/0 |
| RTR-Edge-01 G0/0 | 203.0.113.2/30 | ISP G0/0 |
| RTR-Edge-01 G0/2 | 10.10.254.1/30 | Core G0/1 |
| SW-CORE-01 G0/1 | 10.10.254.2/30 | Edge G0/2 |
| RTR-ISP-01 G0/1 | 198.51.100.1/24 | External LAN |

A /24 uses 255.255.255.0 and a /30 uses 255.255.255.252. I used 198.51.100.0/24 for the simulated external LAN and 203.0.113.0/30 for the ISP-to-edge link.

### Routing decisions

| Device | Route | Next hop |
|---|---|---|
| Core | Default 0.0.0.0/0 | 10.10.254.1 |
| Edge | Office summary 10.10.0.0/16 | 10.10.254.2 |
| Edge | Default 0.0.0.0/0 | 203.0.113.1 |
| ISP | Directly connected networks | No static route required |

The core sends external traffic to the edge. The edge has a return route to the office subnets and a default route to the ISP.

# Endpoint addressing and DHCP plan

I used static addresses for servers, printers, management interfaces and the admin PC. Normal client PCs used DHCP. This kept the devices referenced by ACLs and DNS at predictable addresses.

| Device | Static IPv4 address | Default gateway |
|---|---|---|
| SRV-Infra-01 | 10.10.70.10 | 10.10.70.1 |
| SRV-Intra-01 | 10.10.70.20 | 10.10.70.1 |
| WEB-DMZ-01 | 172.16.50.10 | 172.16.50.1 |
| SRV-Internet-01 | 198.51.100.10 | 198.51.100.1 |
| PC-EXT-TEST-01 | 198.51.100.20 | 198.51.100.1 |
| PC-ADMIN-01 | 10.10.30.10 | 10.10.30.1 |
| Office printer | 10.10.40.10 | 10.10.40.1 |
| Finance printer | 10.10.40.11 | 10.10.40.1 |
| SW-ACC-01 management | 10.10.99.11 | 10.10.99.1 |
| SW-ACC-02 management | 10.10.99.12 | 10.10.99.1 |

All addresses in this table use a /24 mask. Internal clients use DNS 10.10.70.10. The guest and external PC use DNS 198.51.100.10. The DMZ and ISP switches do not have management IPs in this design.

## DHCP pools on SRV Infra 01

| Pool | Address range | Max clients | DNS |
|---|---|---|---|
| POOL-STAFF | 10.10.10.100 to 10.10.10.199 | 100 | 10.10.70.10 |
| POOL-FINANCE | 10.10.20.100 to 10.10.20.149 | 50 | 10.10.70.10 |
| POOL-IT | 10.10.30.100 to 10.10.30.149 | 50 | 10.10.70.10 |
| POOL-GUEST | 10.10.60.100 to 10.10.60.199 | 100 | 198.51.100.10 |

Each pool uses the corresponding VLAN gateway ending in .1 and a /24 mask. The core relays requests from VLANs 10, 20, 30 and 60 to 10.10.70.10. I kept the admin PC at 10.10.30.10, outside the IT DHCP pool.

| Pool Name | Default Gateway | DNS Server | Start IP Address | Subnet Mask | Max User | TFTP Server | WLC Address |
|---|---|---|---|---|---|---|---|
| POOL-GUEST | 10.10.60.1 | 198.51.... | 10.10.6... | 255.255... | 100 | 0.0.0.0 | 0.0.0.0 |
| POOL-IT | 10.10.30.1 | 10.10.7... | 10.10.3... | 255.255... | 50 | 0.0.0.0 | 0.0.0.0 |
| POOL-FINANCE | 10.10.20.1 | 10.10.7... | 10.10.2... | 255.255... | 50 | 0.0.0.0 | 0.0.0.0 |
| POOL-STAFF | 10.10.10.1 | 10.10.7... | 10.10.1... | 255.255... | 100 | 0.0.0.0 | 0.0.0.0 |
| serverPool | 0.0.0.0 | 0.0.0.0 | 10.10.70.0 | 255.255... | 512 | 0.0.0.0 | 0.0.0.0 |

**Figure 2 DHCP pools on SRV-Infra-01.**

# Building the simulated ISP

I started with the ISP section so I had an external network to test against before configuring the office. I assigned addresses to the ISP router, external server and external PC, then enabled HTTP and DNS on SRV-Internet-01.

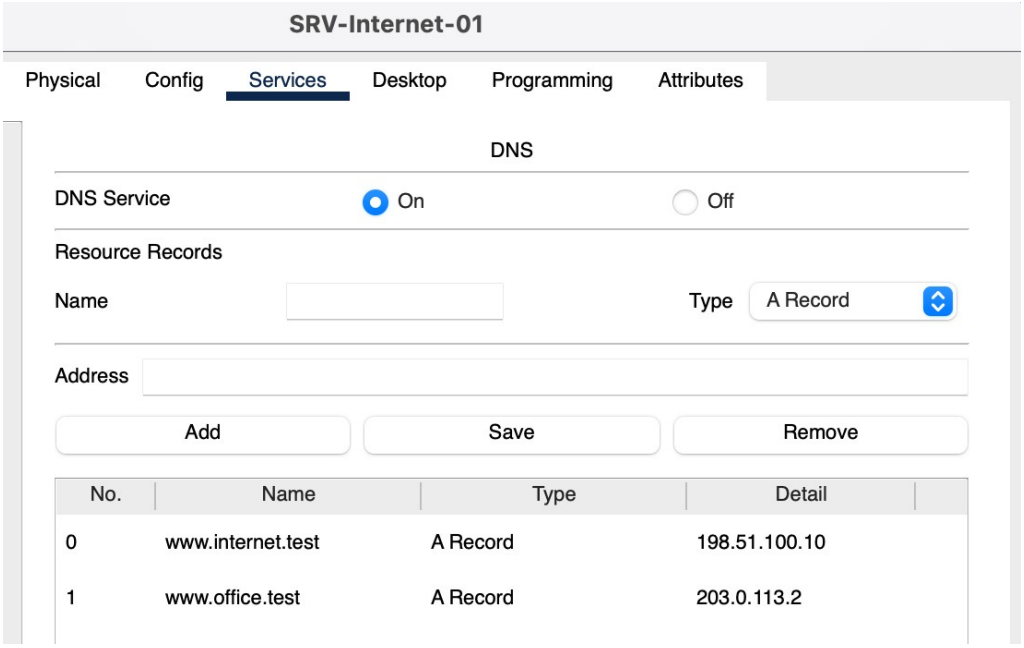



**Figure 3 External DNS records: www.internet.test points to 198.51.100.10 and www.office.test points to 203.0.113.2.**

The external devices share the 198.51.100.0/24 LAN through SW-ISP-01. The ISP router connects that LAN to the office WAN link. The external PC uses 198.51.100.1 as its gateway.

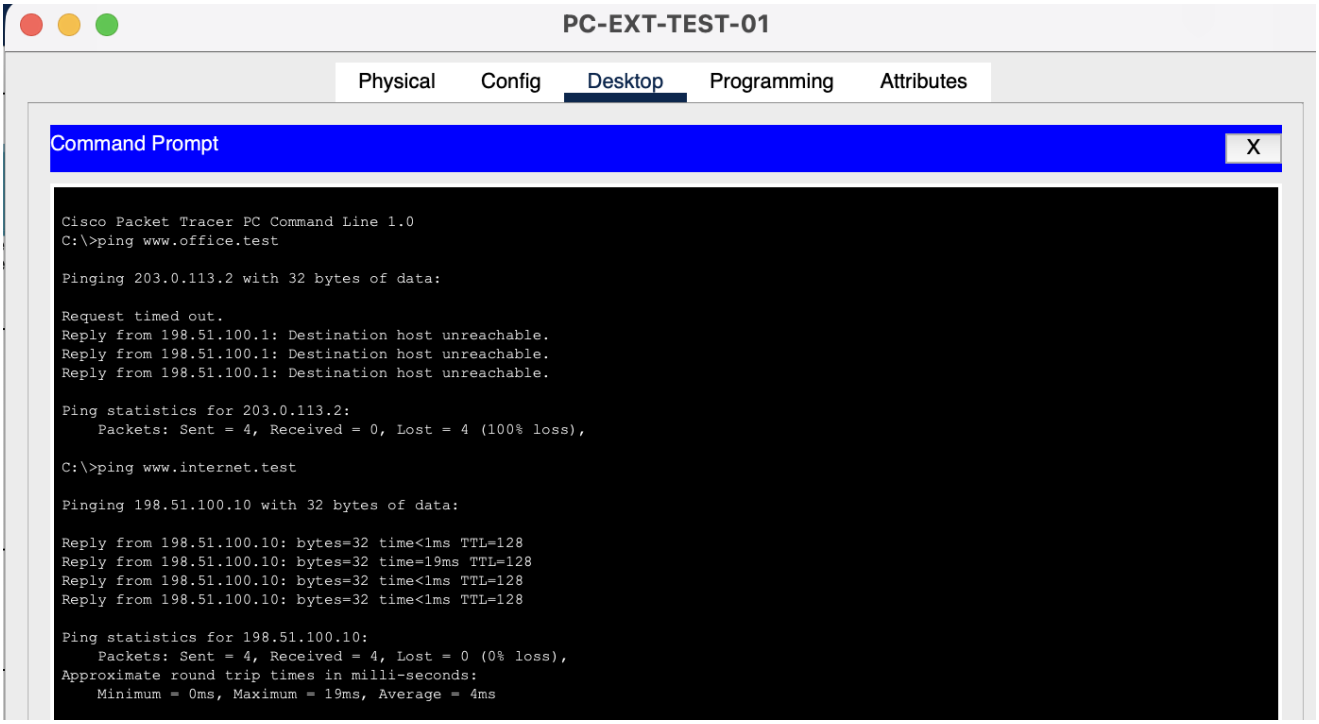



**Figure 4 Early external-PC test: the internet server responds, while the office WAN address is not yet reachable.**

The external PC could resolve both website names and ping the internet server. The office address was still unreachable because I had not finished configuring the edge and DMZ.

# Publishing the DMZ website

I connected the DMZ to its own edge-router interface and placed WEB-DMZ-01 on VLAN 50. The server uses 172.16.50.10 with gateway 172.16.50.1. I enabled HTTP and changed the page heading so I could recognise it during testing.

```
ip nat inside source static tcp 172.16.50.10 80 203.0.113.2 80
```

This mapping forwards requests arriving on 203.0.113.2 at TCP port 80 to the DMZ web server. I set G0/0 as NAT outside and G0/1 as NAT inside.

```
RTR-Edge-01#show ip interf brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     203.0.113.2     YES manual up                    up
GigabitEthernet0/1     172.16.50.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
RTR-Edge-01#ping 203.0.113.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

RTR-Edge-01#ping 172.16.50.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.50.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 9/14/28 ms

RTR-Edge-01#ping 172.16.50.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.50.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
```

**Figure 5 WAN and DMZ interface checks before connecting the core.**

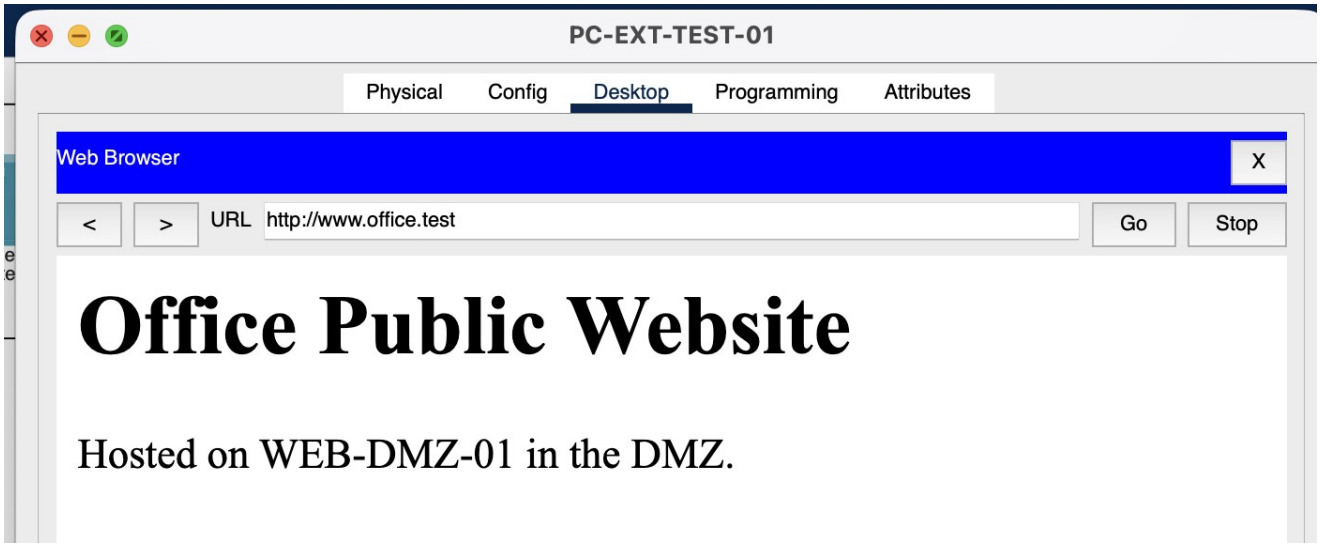



**Figure 6 The external PC successfully opens http://www.office.test after the DMZ NAT role was corrected.**

The website initially timed out because I had set G0/1 to NAT outside. After changing it to NAT inside, I could open the page from the external PC.

# Configuring the core and VLAN trunks

I enabled ip routing on the core, created the VLAN gateway interfaces and made G0/1 a routed port to the edge. ACC-01 carries staff, Finance, printers and management. ACC-02 carries IT, printers, guests and management.

```
interface Vlan10
 ip address 10.10.10.1 255.255.255.0
 ip helper-address 10.10.70.10
 ip access-group STAFF-IN in
!
interface FastEthernet0/23
 switchport trunk allowed vlan 10,20,40,99
 switchport trunk encapsulation dot1q
 switchport mode trunk
```

VLAN 10 uses a helper address to forward DHCP requests to the infrastructure server. I added STAFF-IN after checking basic connectivity. VLAN 40 is allowed on both access-switch trunks because each switch has a printer connected.

```
SW-CORE-01#ping 10.10.254.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.254.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

SW-CORE-01#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.10.254.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.10.10.0/24 is directly connected, Vlan10
C       10.10.20.0/24 is directly connected, Vlan20
C       10.10.30.0/24 is directly connected, Vlan30
C       10.10.40.0/24 is directly connected, Vlan40
C       10.10.60.0/24 is directly connected, Vlan60
C       10.10.70.0/24 is directly connected, Vlan70
C       10.10.99.0/24 is directly connected, Vlan99
C       10.10.254.0/30 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 10.10.254.1
```

**Figure 7 Core routing table showing the internal connected networks and default route through 10.10.254.1.**

When the admin PC could not reach its gateway, I checked its address, the VLAN membership and trunk settings. The PC was connected to ACC-02 F0/3, which was in VLAN 1. Moving it to F0/2 in VLAN 30 restored access to the gateway and management addresses.

# Checking client addressing

After setting up the DHCP pools and relay, I ran ipconfig /all on each client. I checked that its address, gateway and DNS server matched its VLAN.

| Client screenshot | Leased address | Gateway | DNS |
|---|---|---|---|
| PC-STAFF-01 | 10.10.10.100 | 10.10.10.1 | 10.10.70.10 |
| PC-FIN-01 | 10.10.20.100 | 10.10.20.1 | 10.10.70.10 |
| PC-IT-01 | 10.10.30.100 | 10.10.30.1 | 10.10.70.10 |
| PC-GUEST-01 | 10.10.60.100 | 10.10.60.1 | 198.51.100.10 |

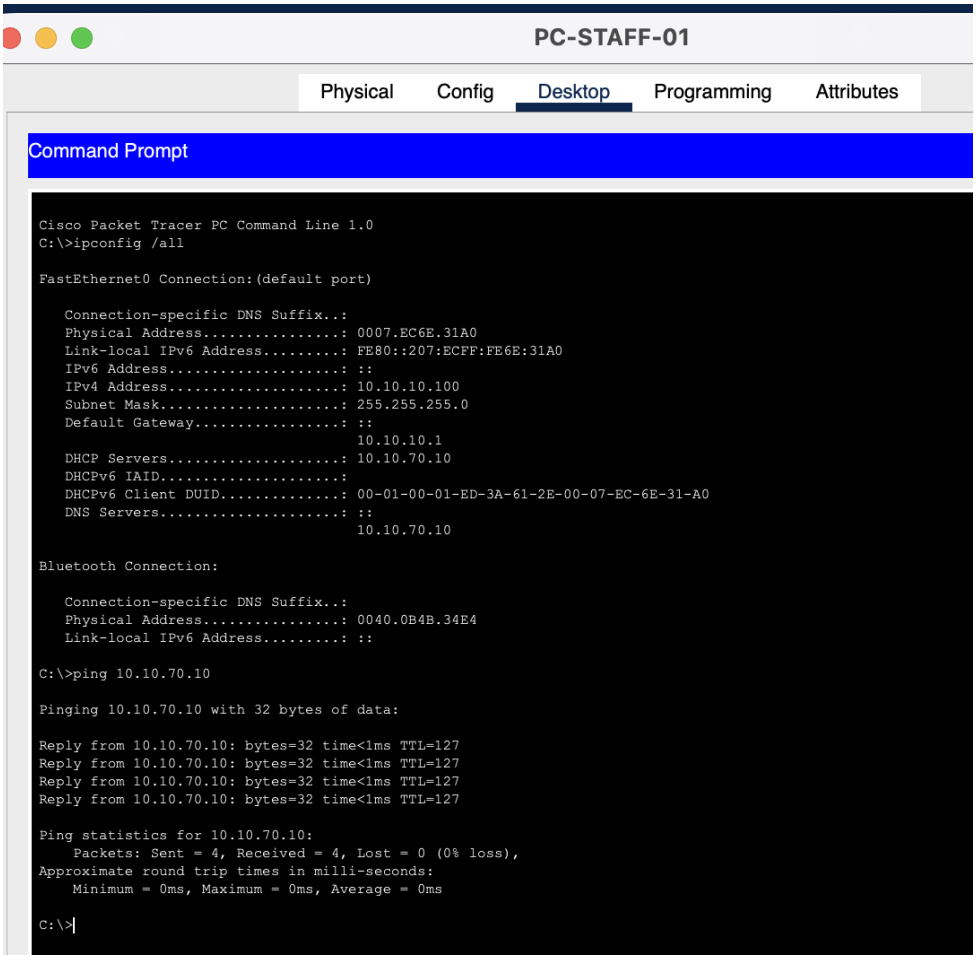



**Figure 8 Staff DHCP lease and infrastructure-server ping before applying ACLs.**

I took this ping test before adding the ACLs. Staff later kept access to DNS, but server pings were restricted to the admin PC. The guest PC also renewed its DHCP lease successfully.

# Internal DNS and website access

I used different DNS answers for the public website depending on where the client was located. External DNS points www.office.test to 203.0.113.2. Internal DNS points the same name directly to 172.16.50.10, so internal clients do not depend on NAT loopback.

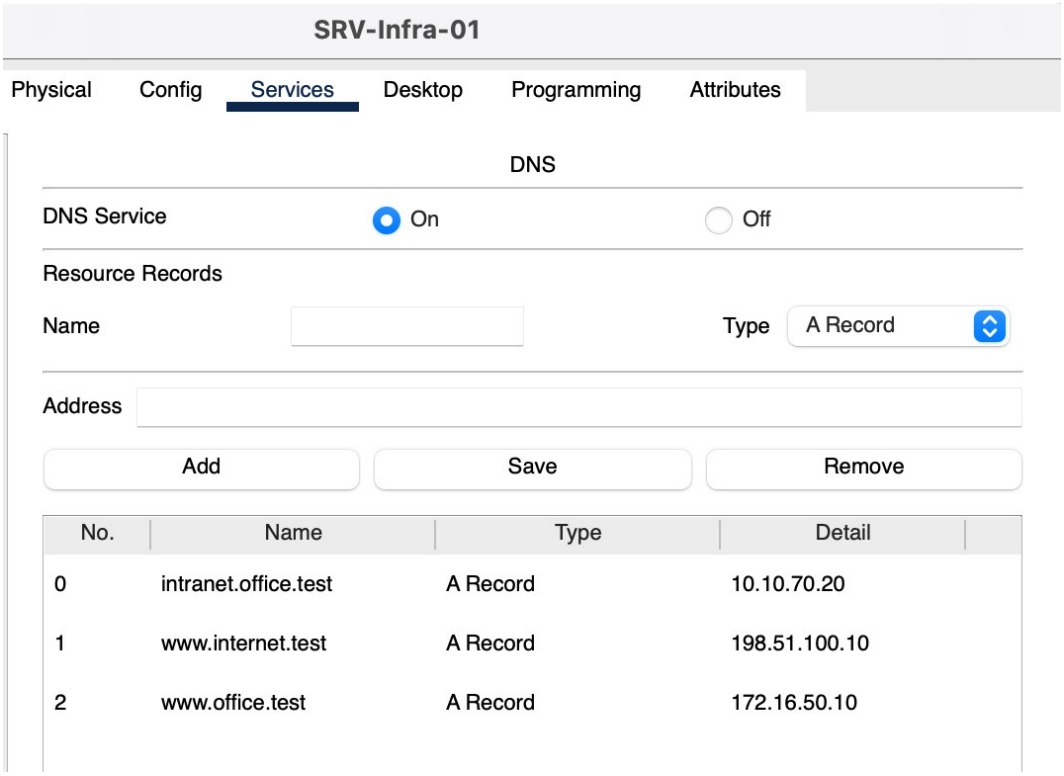



**Figure 9 Internal DNS records for the intranet, simulated internet website and DMZ website.**

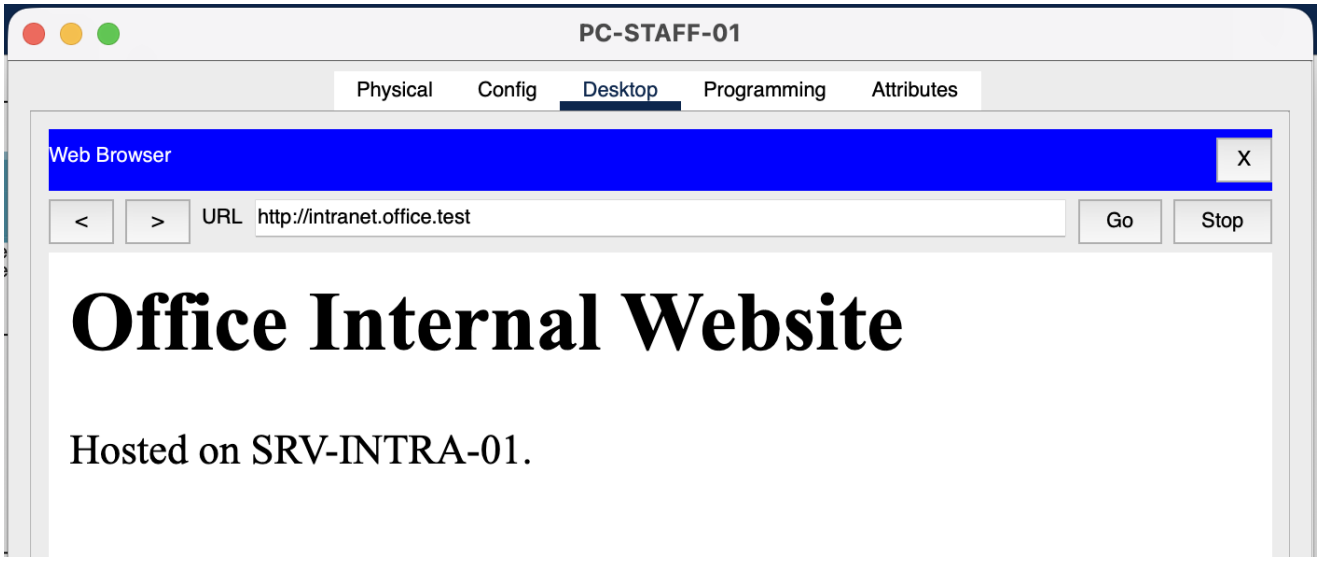



**Figure 10 A staff PC opens the internal website using intranet.office.test.**

SRV-Intra-01 serves the internal website from 10.10.70.20. It stays in VLAN 70 with the infrastructure server, while the public website stays in the DMZ. This separates the published service from the internal service network. From a staff PC, I could also open www.office.test on the DMZ server.

# Adding outbound PAT

Once internal connectivity worked, I added PAT to the edge for VLANs 10, 20, 30 and 60. These clients share 203.0.113.2 for outbound traffic. The server, printer and management subnets are not included in the PAT selection ACL.

```
ip access-list standard PAT-CLIENTS
 permit 10.10.10.0 0.0.0.255
 permit 10.10.20.0 0.0.0.255
 permit 10.10.30.0 0.0.0.255
 permit 10.10.60.0 0.0.0.255
!
ip nat inside source list PAT-CLIENTS interface GigabitEthernet0/0 overload
```

G0/2, facing the core, is NAT inside. The existing static HTTP mapping for the DMZ stays in place alongside PAT. The PAT ACL selects traffic for translation; the interface ACLs enforce access restrictions.

```
Dynamic mappings:
RTR-Edge-01#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
udp 203.0.113.2:1025   10.10.60.100:1025  198.51.100.10:53   198.51.100.10:53
tcp 203.0.113.2:1025   10.10.60.100:1025  198.51.100.10:80   198.51.100.10:80
tcp 203.0.113.2:1027   10.10.10.100:1027  198.51.100.10:80   198.51.100.10:80
tcp 203.0.113.2:80     172.16.50.10:80    ---                ---
tcp 203.0.113.2:80     172.16.50.10:80    198.51.100.20:1035 198.51.100.20:1035
```

**Figure 11  NAT translations showing staff and guest traffic sharing the WAN address, alongside the DMZ TCP 80 mapping.**

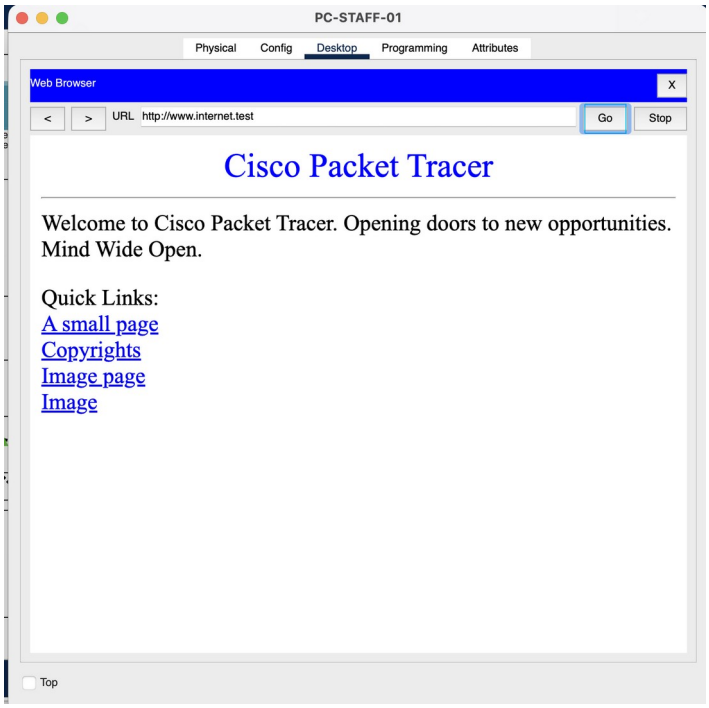



**Figure 12  A staff PC reaches the simulated internet website after PAT is configured.**

I opened the external website from a staff PC, then checked show ip nat translations. The table showed staff and guest traffic using 203.0.113.2 with different port mappings.

# Applying client and perimeter ACLs

I applied the client ACLs inbound on their core VLAN interfaces. Each list permits the required services before denying other private destinations, then allows selected external traffic. The edge filters traffic arriving from the ISP and DMZ.

| ACL | Attachment | Main purpose |
|---|---|---|
| STAFF-IN | Core Vlan10 inbound | DNS, intranet, office printer, DMZ HTTP and external web; block other private access |
| FINANCE-IN | Core Vlan20 inbound | Same service access, with both printers permitted |
| IT-IN | Core Vlan30 inbound | Normal IT service access; admin .30.10 has a routed-access exception |
| GUEST-IN | Core Vlan60 inbound | DHCP, external DNS and web; block other private destinations except own-gateway ping |
| WAN-IN | Edge G0/0 inbound | Published HTTP and selected web, DNS and ICMP return traffic |
| DMZ-IN | Edge G0/1 inbound | Web-server HTTP replies and echo replies; deny other traffic |

I used 203.0.113.2 as the destination in WAN-IN because the inbound WAN ACL sees the public address before the destination is translated. The final list places the ICMP type 11 permit before deny ip any any.

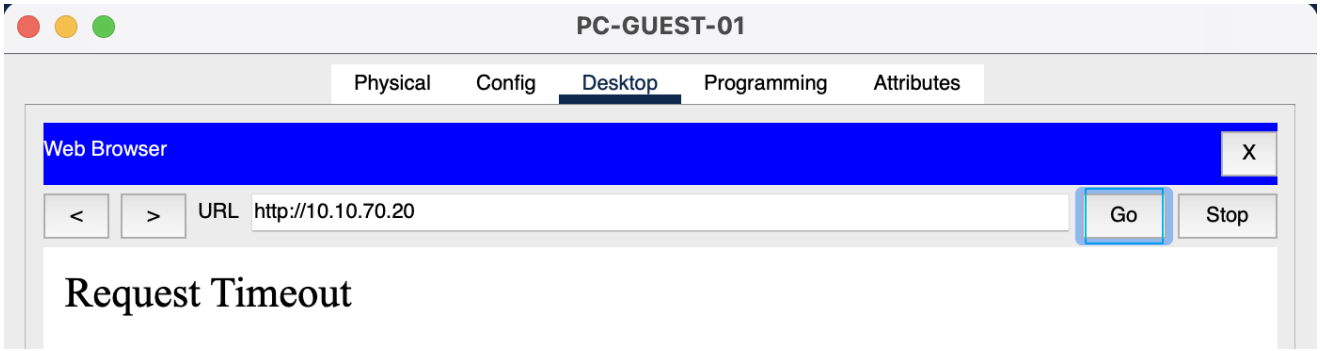



**Figure 13 The guest PC cannot open the internal website at 10.10.70.20.**

I tested allowed and blocked traffic separately. Staff could reach the office printer but not the Finance printer; Finance could reach both. Guests could use the external website but not the intranet or the direct DMZ address. The external PC could still open the office website even though external ping to the edge was blocked. Staff could not ping a Finance PC, and normal IT could not ping the management addresses. The admin PC could ping both access-switch management addresses. The DMZ server could not ping the infrastructure server.

# Restricting servers and printers

After the client and perimeter rules were working, I restricted traffic entering routing from the server and printer VLANs. I allowed the response traffic needed by the lab services, then denied other routed traffic from those sources.

| ACL | Attachment | Permitted traffic |
|---|---|---|
| SERVERS-IN | Core Vlan70 inbound | DHCP replies to relays and clients; DNS replies; intranet HTTP replies; server echo replies to admin |
| PRINTERS-IN | Core Vlan40 inbound | TCP replies with ACK/RST flags and ICMP echo replies to internal addresses |

### Server rules that preserve DHCP and DNS

```
ip access-list extended SERVERS-IN
 permit udp host 10.10.70.10 eq bootps 10.10.0.0 0.0.255.255 eq bootps
 permit udp host 10.10.70.10 eq bootps any eq bootpc
 permit udp host 10.10.70.10 eq domain 10.10.0.0 0.0.255.255
 permit tcp host 10.10.70.10 eq domain 10.10.0.0 0.0.255.255 established
```

The first two rules cover replies to the relay and replies directly to DHCP clients. The rest of the final ACL permits intranet HTTP responses and diagnostic echo replies to 10.10.30.10, followed by a deny for other traffic.

### Testing the direction of traffic

After applying these rules, I checked that the admin PC could still ping the servers, staff could load the intranet, and Finance could reach both printers. A ping started from the infrastructure server to the admin PC was blocked.

| Test | Result |
|---|---|
| Admin to infrastructure and intranet servers | Allowed |
| Infrastructure server to admin PC | Blocked |
| Staff to intranet | Allowed |
| Staff to office printer | Allowed |
| Staff to Finance printer | Blocked |
| Finance and admin to both printers | Allowed |

The client ACLs control access to each printer. PRINTERS-IN allows the replies back to the clients.

# SSH management and unused ports

I enabled SSH version 2 on the edge, core and both office access switches. I created the local netadmin account, configured an enable secret, and applied ADMIN-SSH to the VTY lines to permit only 10.10.30.10. Telnet is excluded by transport input ssh. Console login also uses the local account.

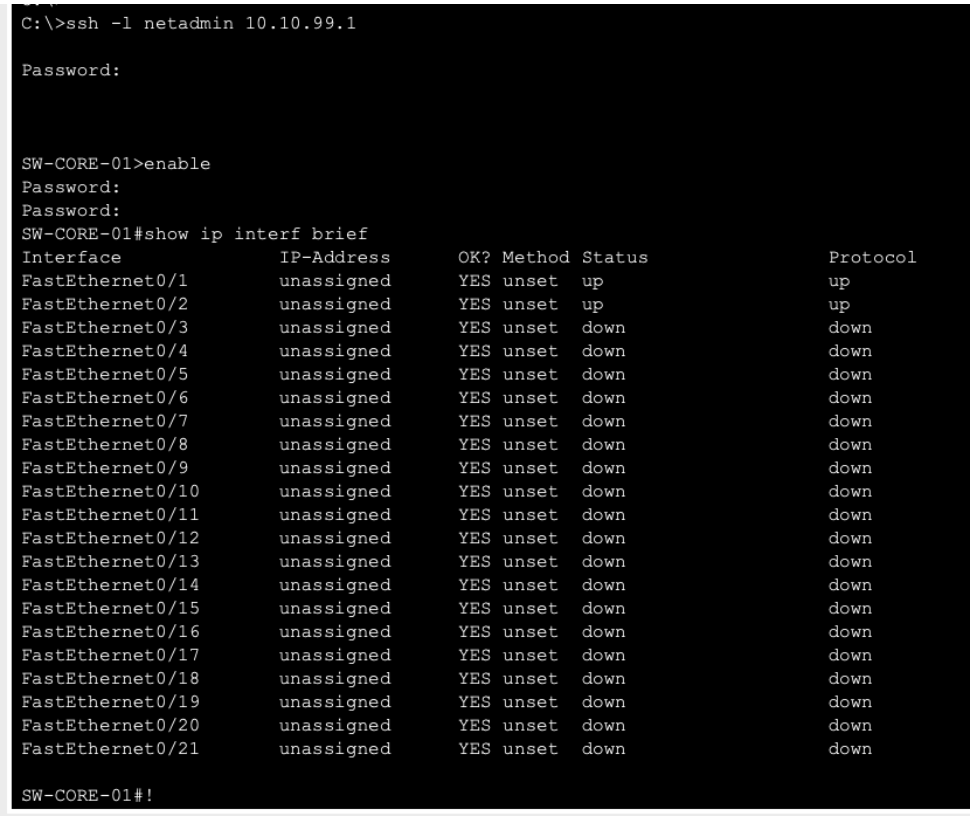



**Figure 14 Admin SSH login and enable access to the core, before shutting down unused ports.**

| Managed device | SSH address |
|---|---|
| RTR-Edge-01 | 10.10.254.1 |
| SW-CORE-01 | 10.10.99.1 |
| SW-ACC-01 | 10.10.99.11 |
| SW-ACC-02 | 10.10.99.12 |

I could log in over SSH from the admin PC to all four devices. Telnet sessions were closed, and SSH attempts from other PCs failed.

## Unused ports

On the core, F0/3 to F0/22 and G0/2 are shut down in VLAN 999. On ACC-01, the unused ports are F0/4 to F0/12, F0/15 to F0/20, F0/22 to F0/24 and G0/2. On ACC-02, they are F0/3 to F0/4, F0/6 to F0/8, F0/10 to F0/24 and G0/2.

# Troubleshooting during the build

### External website requests timed out

The edge router could ping 172.16.50.10, and the external PC could ping 203.0.113.2, but the office webpage did not load. I checked show ip nat statistics and found both G0/0 and G0/1 listed as outside interfaces, with no inside interface. I removed ip nat outside from G0/1 and applied ip nat inside. After that, the external PC could open the website.

### The admin PC could not reach its gateway

The admin PC had 10.10.30.10/24 and gateway 10.10.30.1, but could not ping its gateway. I checked the VLANs and trunks, then noticed the PC was connected to ACC-02 F0/3 instead of F0/2. F0/3 was in VLAN 1. Moving the cable to F0/2 in VLAN 30 fixed the connection.

### Checking DNS settings

I tested the public website by IP address and DNS name to narrow down the connection problem. I also corrected the admin PC DNS address from 10.10.70.1 to 10.10.70.10.

### Some ACL test results did not match the policy

Some early tests did not match the restrictions I wanted: staff could reach Finance resources, and normal IT could reach management addresses. I checked the ACL entries, their interface attachments and match counters, then reapplied the intended attachments and tested again. The connections were then allowed or blocked as expected.

### An ICMP keyword was not accepted

Packet Tracer rejected time-exceeded in the WAN ACL, so I used ICMP type 11. I also caught that the permit was below deny ip any any. I moved the deny to the end so traffic could reach the permit rule.

### Retesting the connections

For the final checks, I used browser tests for websites, SSH sessions for management access and ping for reachability. I also compared ACL counters before and after a blocked test to see which rule was matching.

# Final printer access tests

From a staff PC, I could ping the office printer at 10.10.40.10 but not the Finance printer at 10.10.40.11. From a Finance PC, both printers replied.

```
C:\>ping 10.10.40.10

Pinging 10.10.40.10 with 32 bytes of data:

Reply from 10.10.40.10: bytes=32 time<1ms TTL=127
Reply from 10.10.40.10: bytes=32 time<1ms TTL=127
Reply from 10.10.40.10: bytes=32 time=20ms TTL=127
Reply from 10.10.40.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 20ms, Average = 5ms

C:\>ping 10.10.40.11

Pinging 10.10.40.11 with 32 bytes of data:

Reply from 10.10.10.1: Destination host unreachable.
Reply from 10.10.10.1: Destination host unreachable.
Reply from 10.10.10.1: Destination host unreachable.
Reply from 10.10.10.1: Destination host unreachable.

Ping statistics for 10.10.40.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

**Figure 15  Staff PC: office printer ping succeeds; Finance printer ping fails with destination-unreachable messages from 10.10.10.1.**

```
C:\>ping 10.10.40.10

Pinging 10.10.40.10 with 32 bytes of data:

Reply from 10.10.40.10: bytes=32 time<1ms TTL=127
Reply from 10.10.40.10: bytes=32 time<1ms TTL=127
Reply from 10.10.40.10: bytes=32 time<1ms TTL=127
Reply from 10.10.40.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.40.11

Pinging 10.10.40.11 with 32 bytes of data:

Reply from 10.10.40.11: bytes=32 time<1ms TTL=127
Reply from 10.10.40.11: bytes=32 time<1ms TTL=127
Reply from 10.10.40.11: bytes=32 time=22ms TTL=127
Reply from 10.10.40.11: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.40.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 22ms, Average = 5ms
```

**Figure 16  Finance PC: all four ping replies are received from each printer.**

# Final guest access tests

The guest PC could open www.internet.test. Requests to the intranet at 10.10.70.20 and the DMZ website at 172.16.50.10 timed out, as intended.

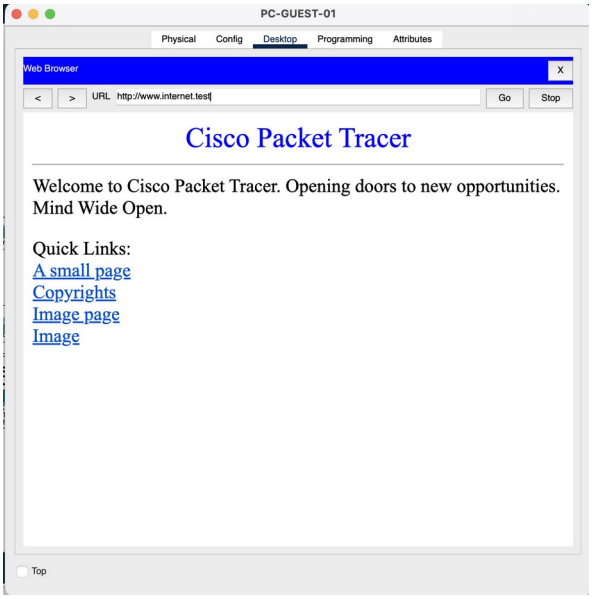



**Figure 17 PC-GUEST-01 loads http://www.internet.test.**

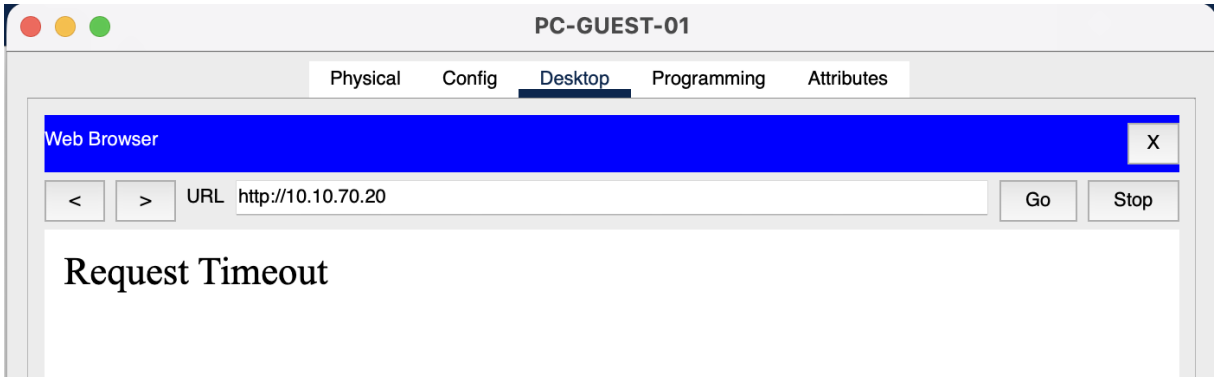



**Figure 18 The guest request to http://10.10.70.20 times out.**

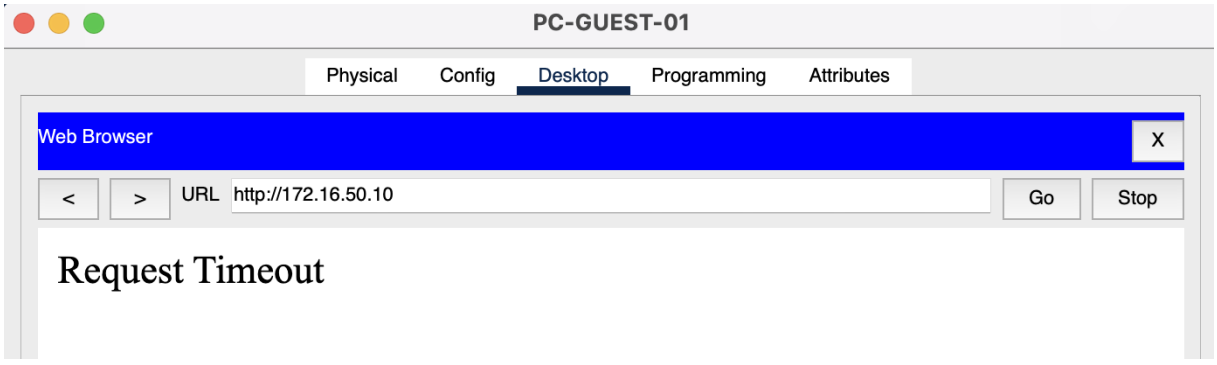



**Figure 19 The guest request to http://172.16.50.10 times out.**

# Final external access tests

The external test PC could open www.office.test, while ping to the edge WAN address was blocked. WAN-IN permits the published HTTP service and blocks incoming echo requests.

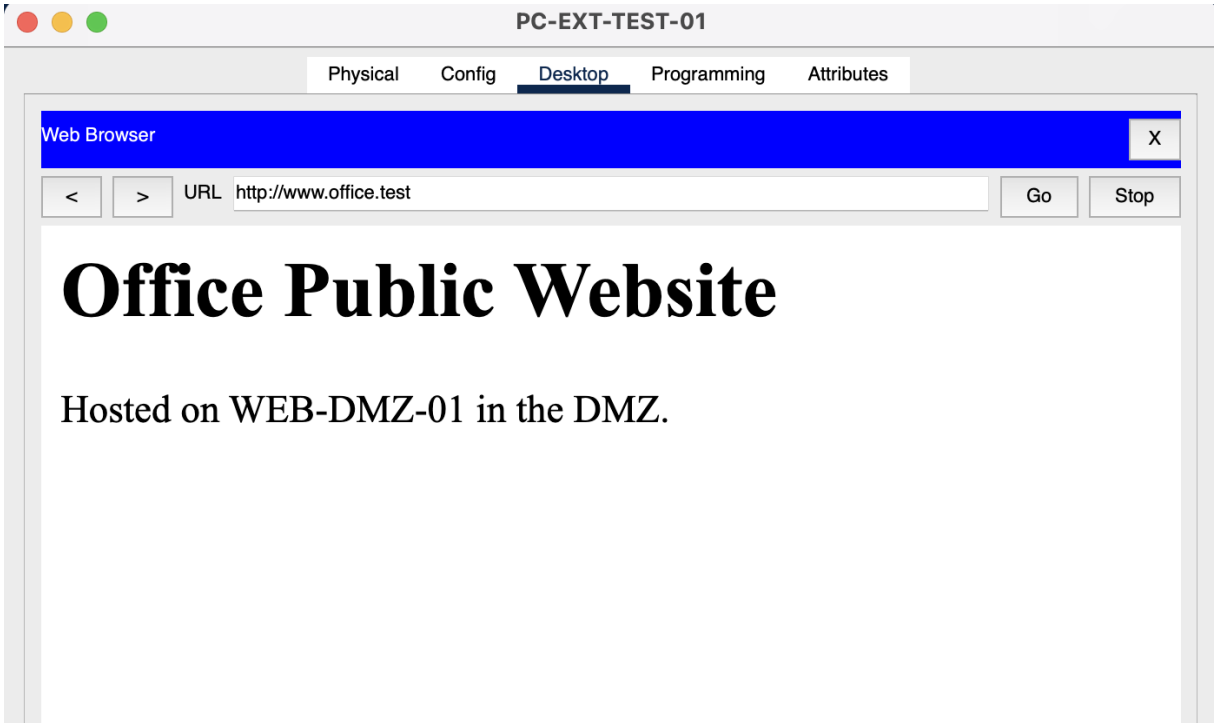



**Figure 20 PC-EXT-TEST-01 loads http://www.office.test and displays the page hosted on WEB-DMZ-01.**

```
C:\>ping 203.0.113.2

Pinging 203.0.113.2 with 32 bytes of data:

Reply from 203.0.113.2: Destination host unreachable.
Reply from 203.0.113.2: Destination host unreachable.
Reply from 203.0.113.2: Destination host unreachable.
Reply from 203.0.113.2: Destination host unreachable.

Ping statistics for 203.0.113.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

**Figure 21 External ping to 203.0.113.2 receives destination-unreachable messages and reports 100 percent packet loss.**

# Final remote management tests

I logged in to SW-ACC-01 over SSH from the admin PC and entered enable mode. A Telnet attempt to the core was closed immediately, and SSH attempts from a non-admin PC timed out.

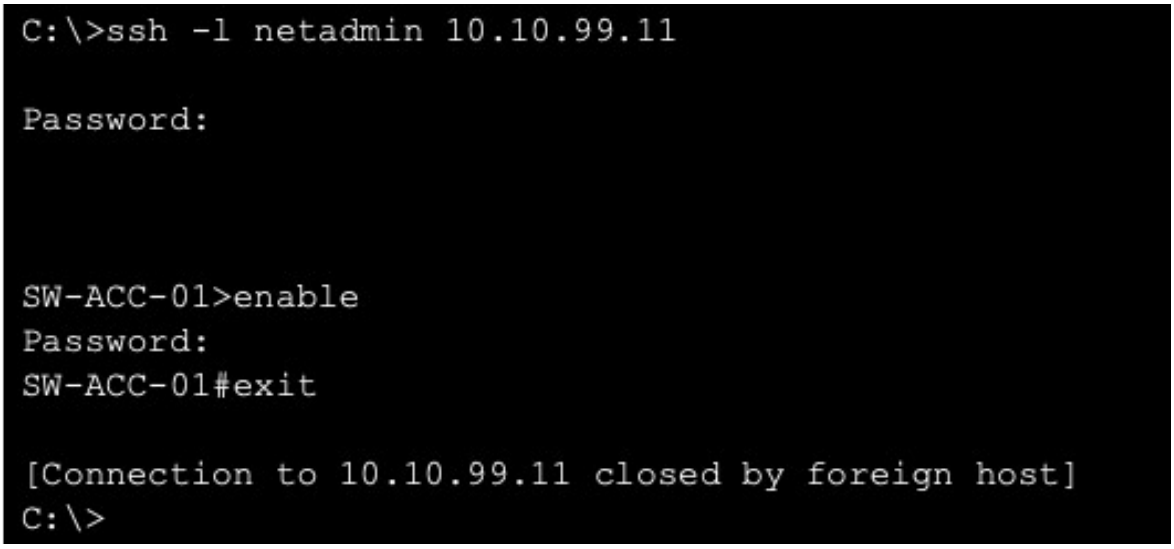



**Figure 22 Admin PC: SSH to SW-ACC-01 at 10.10.99.11 succeeds, followed by successful enable access.**

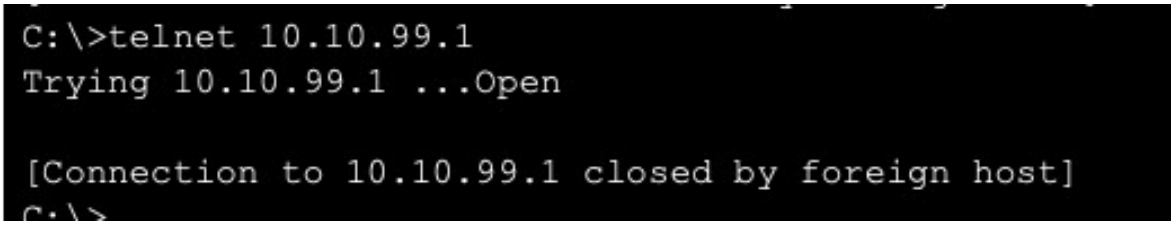



**Figure 23 Telnet to 10.10.99.1 reports Open, then the remote host immediately closes the session.**

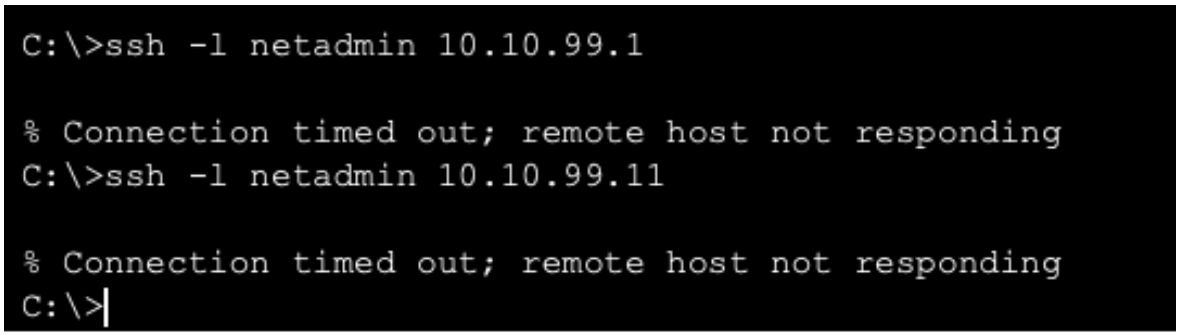



**Figure 24 Non-admin PC: SSH attempts to 10.10.99.1 and 10.10.99.11 time out.**

# Final server access tests

I checked the direction of access after applying SERVERS-IN. The admin PC could ping the infrastructure server, but the infrastructure server could not initiate a ping to the admin PC.

```
C:\>ping 10.10.70.10

Pinging 10.10.70.10 with 32 bytes of data:

Reply from 10.10.70.10: bytes=32 time<1ms TTL=127
Reply from 10.10.70.10: bytes=32 time<1ms TTL=127
Reply from 10.10.70.10: bytes=32 time<1ms TTL=127
Reply from 10.10.70.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.70.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

**Figure 25  Admin PC: ping to SRV-Infra-01 at 10.10.70.10 receives all four replies.**

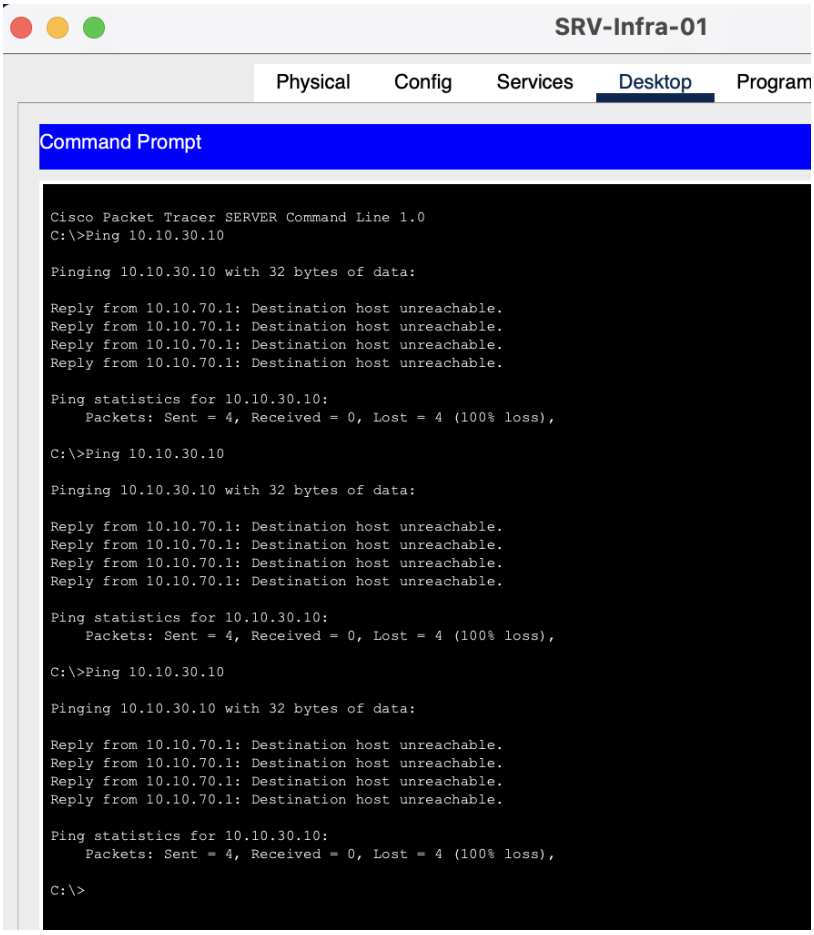



**Figure 26  SRV-Infra-01: repeated pings to admin 10.10.30.10 fail with destination-unreachable messages from 10.10.70.1.**

# Final unused port checks

On SW-ACC-02, I checked both VLAN membership and interface status. The unused ports are in VLAN 999 and administratively down. The active client ports, uplink and management SVI remain up.

```
SW-ACC-02#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
30   IT                               active    Fa0/1, Fa0/2
40   PRINTERS                         active    Fa0/5
60   GUEST                            active    Fa0/9
99   MANAGEMENT                       active
999  UNUSED                           active    Fa0/3, Fa0/4, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

**Figure 27 SW-ACC-02: show vlan brief places unused access ports and G0/2 in VLAN 999.**

```
SW-ACC-02#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual administratively down down
FastEthernet0/4        unassigned      YES manual administratively down down
FastEthernet0/5        unassigned      YES manual up                    up
FastEthernet0/6        unassigned      YES manual administratively down down
FastEthernet0/7        unassigned      YES manual administratively down down
FastEthernet0/8        unassigned      YES manual administratively down down
FastEthernet0/9        unassigned      YES manual up                    up
FastEthernet0/10       unassigned      YES manual administratively down down
FastEthernet0/11       unassigned      YES manual administratively down down
FastEthernet0/12       unassigned      YES manual administratively down down
FastEthernet0/13       unassigned      YES manual administratively down down
FastEthernet0/14       unassigned      YES manual administratively down down
FastEthernet0/15       unassigned      YES manual administratively down down
FastEthernet0/16       unassigned      YES manual administratively down down
FastEthernet0/17       unassigned      YES manual administratively down down
FastEthernet0/18       unassigned      YES manual administratively down down
FastEthernet0/19       unassigned      YES manual administratively down down
FastEthernet0/20       unassigned      YES manual administratively down down
FastEthernet0/21       unassigned      YES manual administratively down down
FastEthernet0/22       unassigned      YES manual administratively down down
FastEthernet0/23       unassigned      YES manual administratively down down
FastEthernet0/24       unassigned      YES manual administratively down down
GigabitEthernet0/1     unassigned      YES manual up                    up
GigabitEthernet0/2     unassigned      YES manual administratively down down
Vlan1                  unassigned      YES manual administratively down down
Vlan99                 10.10.99.12     YES manual up                    up
```

**Figure 28 SW-ACC-02: show ip interface brief confirms the unused ports are administratively down.**

# Checking the matching ACL rule

I ran show access-lists STAFF-IN before and after the blocked printer test. The deny rule for 10.0.0.0/8 increased from 39 to 47 matches, while the other counters stayed the same.

```
SW-CORE-01#
SW-CORE-01#show access-lists STAFF-IN
Extended IP access list STAFF-IN
    permit udp any eq bootpc any eq bootps (6 match(es))
    permit icmp 10.10.10.0 0.0.0.255 host 10.10.10.1 echo
    permit icmp 10.10.10.0 0.0.0.255 host 10.10.30.10 echo-reply
    permit udp 10.10.10.0 0.0.0.255 host 10.10.70.10 eq domain (15 match(es))
    permit tcp 10.10.10.0 0.0.0.255 host 10.10.70.10 eq domain
    permit tcp 10.10.10.0 0.0.0.255 host 10.10.70.20 eq www (25 match(es))
    permit ip 10.10.10.0 0.0.0.255 host 10.10.40.10 (14 match(es))
    permit tcp 10.10.10.0 0.0.0.255 host 172.16.50.10 eq www (20 match(es))
    deny ip any 10.0.0.0 0.255.255.255 (39 match(es))
    deny ip any 172.16.0.0 0.15.255.255
    deny ip any 192.168.0.0 0.0.255.255
    permit tcp 10.10.10.0 0.0.0.255 any eq www (25 match(es))
    permit tcp 10.10.10.0 0.0.0.255 any eq 443
    permit icmp 10.10.10.0 0.0.0.255 any echo
    deny ip any any
```

**Figure 29 Before the test: the deny rule for destinations in 10.0.0.0/8 shows 39 matches.**

```
SW-CORE-01#show access-lists STAFF-IN
Extended IP access list STAFF-IN
    permit udp any eq bootpc any eq bootps (6 match(es))
    permit icmp 10.10.10.0 0.0.0.255 host 10.10.10.1 echo
    permit icmp 10.10.10.0 0.0.0.255 host 10.10.30.10 echo-reply
    permit udp 10.10.10.0 0.0.0.255 host 10.10.70.10 eq domain (15 match(es))
    permit tcp 10.10.10.0 0.0.0.255 host 10.10.70.10 eq domain
    permit tcp 10.10.10.0 0.0.0.255 host 10.10.70.20 eq www (25 match(es))
    permit ip 10.10.10.0 0.0.0.255 host 10.10.40.10 (14 match(es))
    permit tcp 10.10.10.0 0.0.0.255 host 172.16.50.10 eq www (20 match(es))
    deny ip any 10.0.0.0 0.255.255.255 (47 match(es))
    deny ip any 172.16.0.0 0.15.255.255
    deny ip any 192.168.0.0 0.0.255.255
    permit tcp 10.10.10.0 0.0.0.255 any eq www (25 match(es))
    permit tcp 10.10.10.0 0.0.0.255 any eq 443
    permit icmp 10.10.10.0 0.0.0.255 any echo
    deny ip any any
```

**Figure 30 After the test: the same deny rule shows 47 matches. The other displayed counters are unchanged.**

The Finance printer at 10.10.40.11 falls under this deny rule. The office printer at 10.10.40.10 has an earlier permit. This helped me follow the ACL from top to bottom and see why staff could reach one printer but not the other.

# Test results

Select a figure number to open its screenshot. For results recorded in my notes, select the linked keyword in the Test column.

| Test | Result | Reference |
|---|---|---|
| External DNS and internet-server ping | Allowed | Image; Fig 4 |
| External office website after DMZ correction | Allowed | Image; Fig 6 |
| Staff, Finance, IT and guest DHCP addresses | Correct subnets | Fig 8; address table |
| Staff intranet by DNS name | Allowed | Image; Fig 10 |
| Staff DMZ website by DNS name | Allowed | Notes |
| Staff external website after PAT | Allowed | Image; Fig 12 |
| Guest external website | Allowed | Image; Fig 17 |
| Guest DHCP renewal | Allowed | Notes |
| Guest direct intranet access | Blocked | Image; Fig 18 |
| Guest direct DMZ access | Blocked | Image; Fig 19 |
| Staff office printer ping | Allowed | Image; Fig 15 |
| Staff Finance printer ping | Blocked | Image; Fig 15 |
| Staff Finance PC ping | Blocked | Notes |
| Finance both printer pings | Allowed | Image; Fig 16 |
| Normal IT management-address ping | Blocked | Notes |
| Admin both access-switch management pings | Allowed | Notes |
| External edge ping after WAN ACL | Blocked | Image; Fig 21 |
| DMZ to infrastructure-server ping | Blocked | Notes |
| Admin SSH to core | Allowed | Image; Fig 14 |
| Admin SSH to ACC-01 | Allowed | Image; Fig 22 |
| Admin SSH to edge and ACC-02 | Allowed | Notes |
| Admin Telnet session | Closed | Image; Fig 23 |
| Non-admin SSH to core and ACC-01 | Times out | Image; Fig 24 |
| Admin to infrastructure server | Allowed | Image; Fig 25 |
| Admin to intranet server | Allowed | Notes |
| Infrastructure server to admin | Blocked | Image; Fig 26 |
| Intranet and printer retests after final ACLs | Allowed as intended | Notes |

# What I learned and what I would improve

Building the full network helped me understand how the settings depend on each other. A correct IP address did not help when the PC was on the wrong switch port, and a working ping did not mean the website was reachable through NAT.

I got better at narrowing down a fault by checking one part of the connection at a time. Checking the port, VLAN, gateway and NAT settings helped me find the issues in this build. Comparing ACL counters also helped me understand which rule was handling the traffic.

## Lab limitations

I used cisco for the local login and enable secrets in this lab. Both websites use HTTP. I tested printer access with ping.

The network has one core and one edge, so it has no redundancy. The ACLs are stateless: TCP established checks ACK/RST flags, and UDP replies are allowed by address and port. Traffic between devices in the same VLAN bypasses the routed ACLs, including the admin and IT PCs sharing VLAN 30.

## What I would change next

My next improvements would be to put the admin PC in its own VLAN, use stronger credentials, and add central logging. I would also review the extra serverPool shown in the DHCP settings, remove the empty router rip section, and shut down unused ports on the ISP and DMZ switches.

## Project files

| File | Purpose |
|---|---|
| Documentation.pkt | Saved Packet Tracer project |
| Network Build Screenshots.docx | Original screenshots and build notes |
| RTR-EDGE-01.txt | Edge interfaces, routing, NAT, perimeter ACLs and SSH |
| RTR-ISP-01.txt | Simulated ISP router configuration |
| SW-CORE-01.txt | VLAN interfaces, routing, ACLs, SSH and unused ports |
| SW-ACC-01.txt and SW-ACC-02.txt | Office access ports, trunks and management |
| SW-DMZ-01.txt and SW-ISP-01.txt | DMZ and external switching |

I kept the .pkt file alongside the router and switch exports because it also saves the server settings, endpoint addresses and topology.

## Why I built this home lab

I had already built a network topology in a controlled environment using physical equipment. I started this home lab to get a deeper understanding of DMZs, PAT and access restrictions, and to see how those settings work together.

Connecting the office to a simulated internet made me think more about what should be reachable from outside and why. I wanted to make the public website available while keeping internal devices and management access restricted. Testing from different PCs helped me understand why a working connection is only part of the design. I plan to keep developing this project as I learn more and try different ways to secure the network.

# Glossary and navigation

Select a linked term in the report to open its definition. Select the term below to return to its use in the report.

| Term | Meaning in this lab |
|---|---|
| ACL | Access control list. Ordered rules that permit or deny traffic. In this lab, interface ACLs restrict access between networks. |
| DHCP | Dynamic Host Configuration Protocol. Gives clients an IP address, subnet mask, gateway and DNS settings automatically. |
| DHCP relay | Forwards client DHCP requests to a server on another subnet. The core uses ip helper-address to reach 10.10.70.10. |
| DMZ | Demilitarized zone. A separate network for public-facing services. Here, the web server sits apart from internal servers, with ACLs controlling access. |
| DNS | Domain Name System. Resolves names to IP addresses. Internal and external clients receive different addresses for www.office.test. |
| NAT | Network Address Translation. Changes IP addresses in traffic. The static TCP mapping also forwards public port 80 to the DMZ web server. |
| PAT | Port Address Translation. A form of NAT that lets clients share one public IP using distinct port mappings. Configured here with overload. |
| SSH | Secure Shell. Provides an encrypted remote login to network devices. Only the admin PC is allowed to use it in this lab. |
| Stateless ACL | Filters each packet without tracking a full connection. The TCP established keyword checks ACK/RST flags; it is not session tracking. |
| Trunk | A switch link that carries multiple VLANs. Allowed-VLAN lists limit which VLANs can cross the office uplinks. |
| VLAN | Virtual local area network. Separates a switched network into broadcast domains. Routing and ACLs control traffic between the VLANs here. |

### Jump to a section

Addressing | DMZ website | PAT | Access restrictions

SSH and unused ports | Troubleshooting | Test results | Why I built this lab